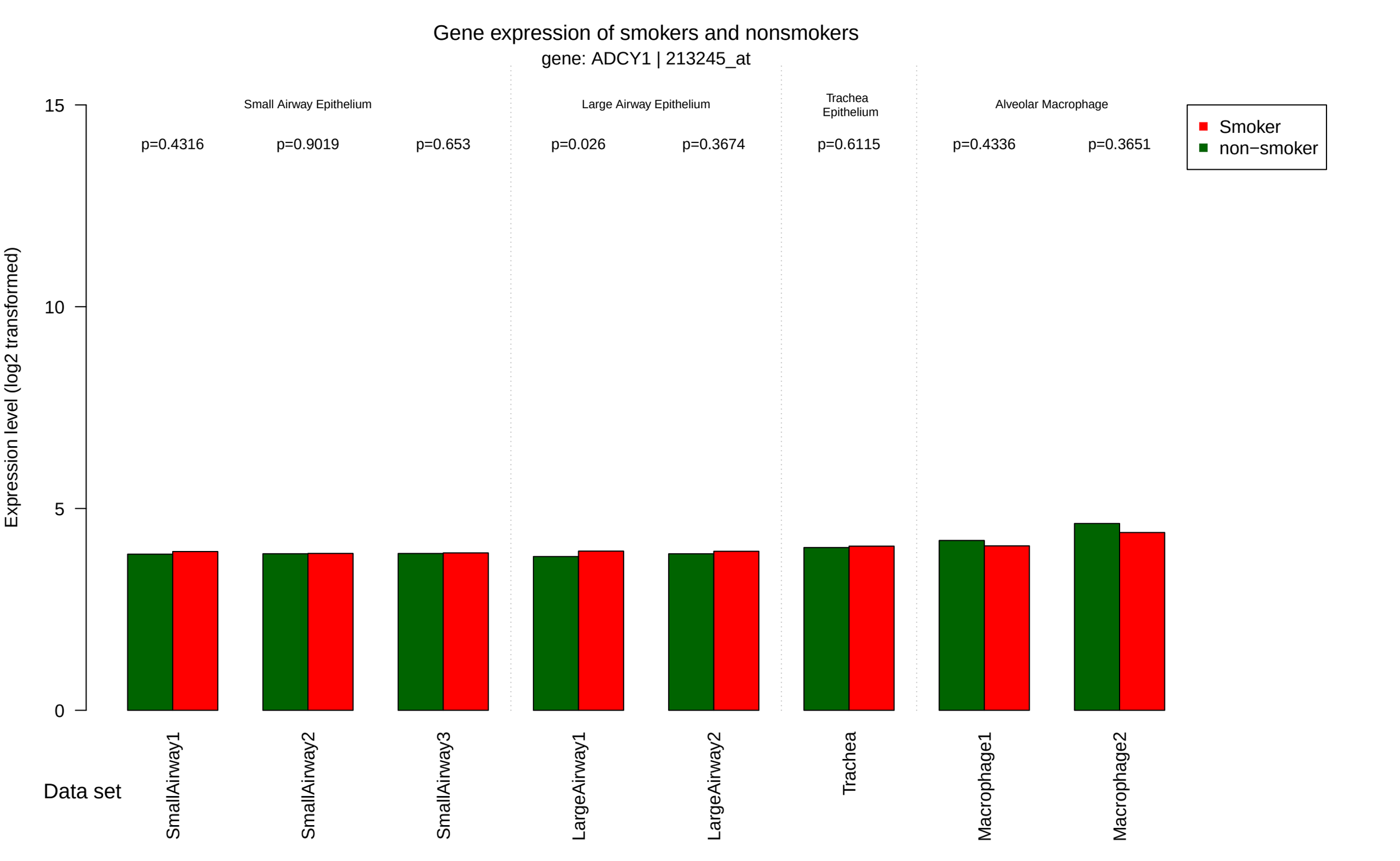
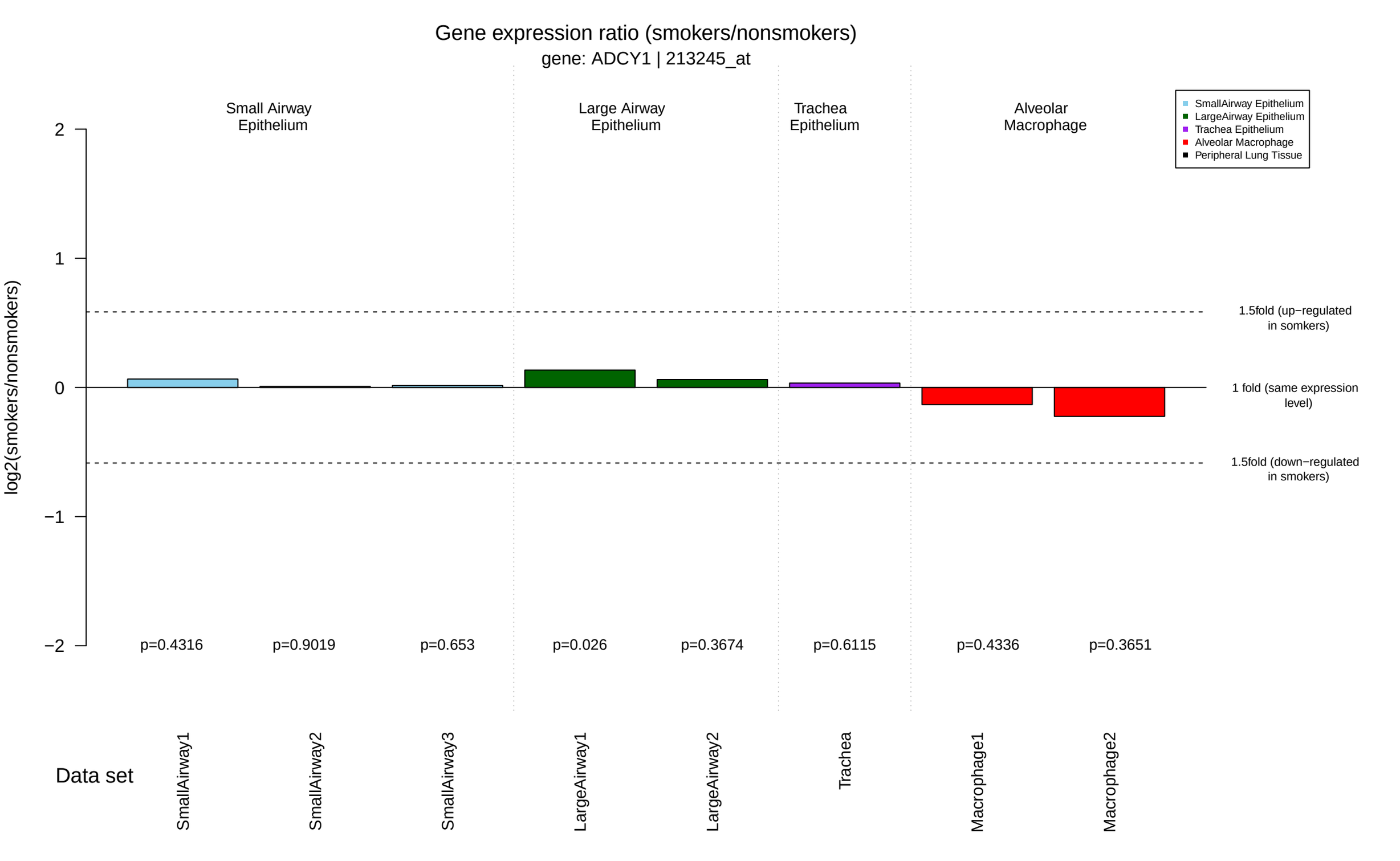
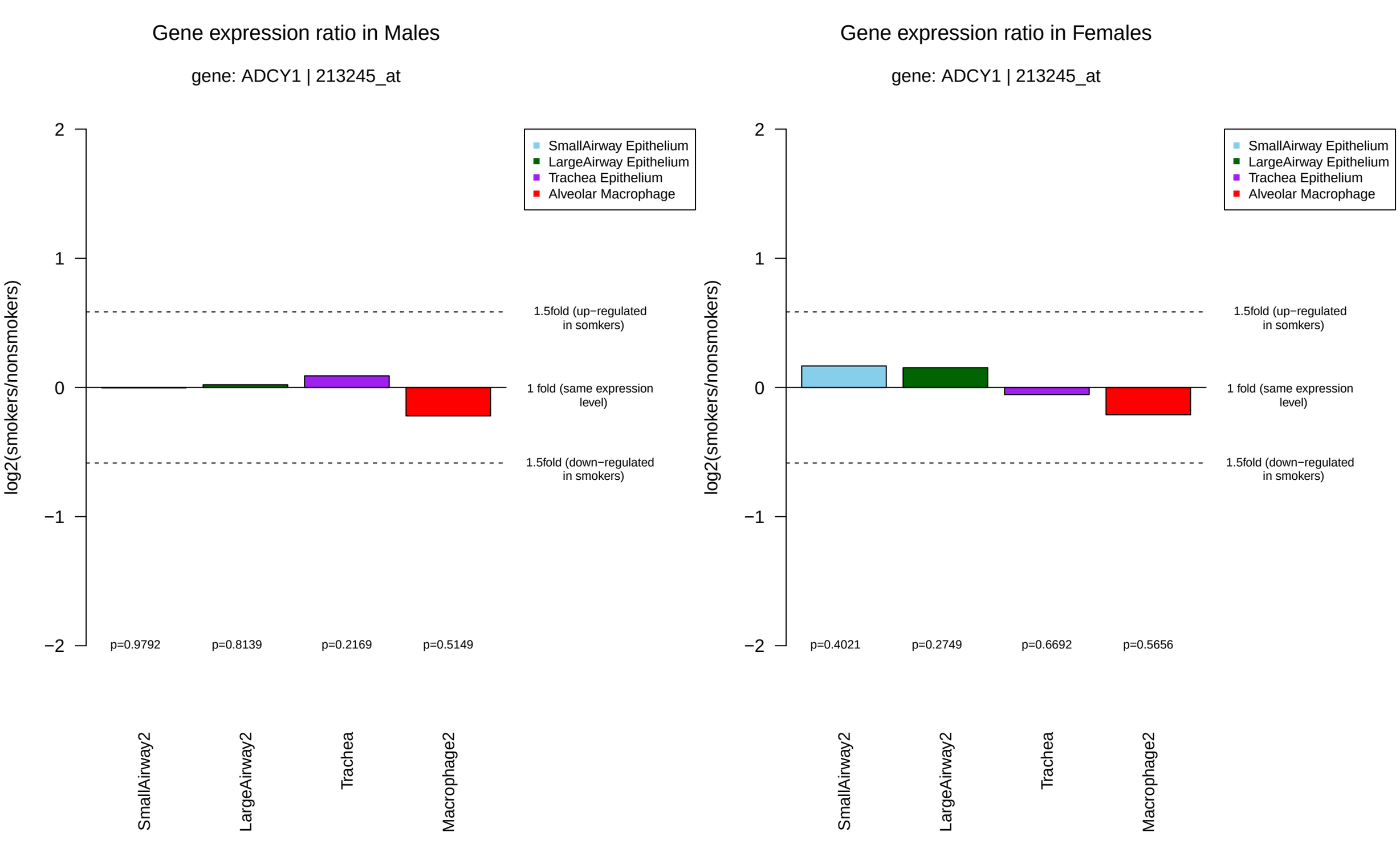
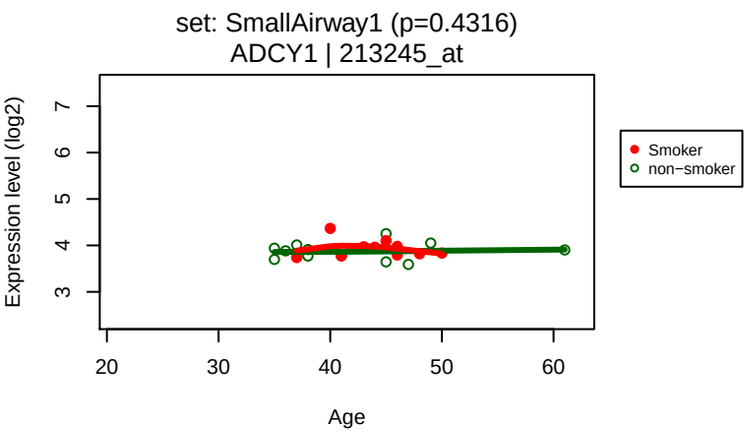
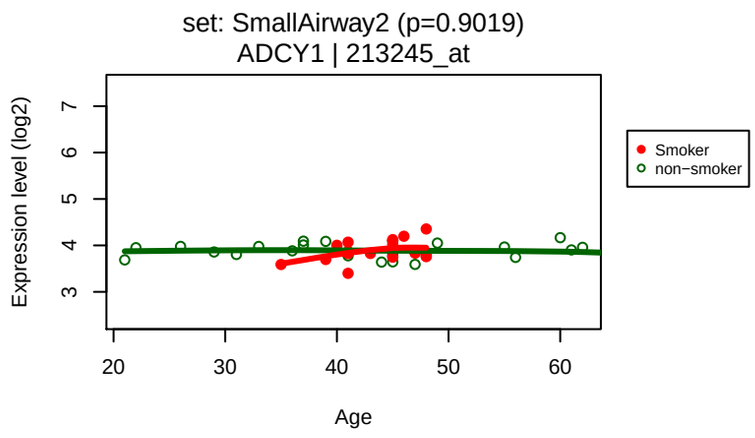
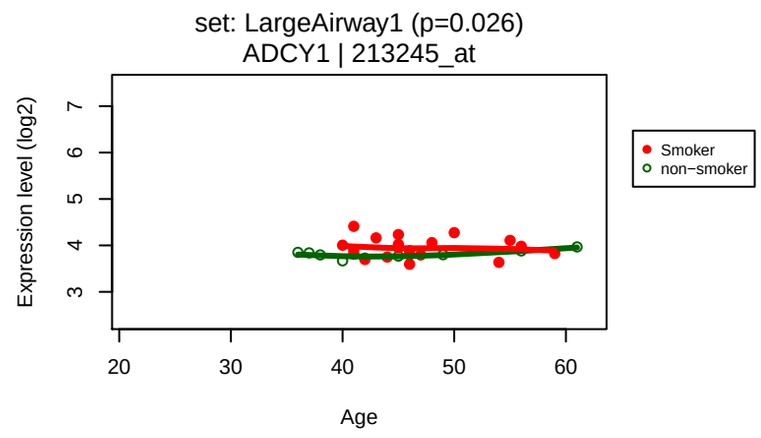

Gene expression of smokers and nonsmokers
gene: ADCY1 | 213245_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.4316
p=0.9019
p=0.653
p=0.026
p=0.3674
p=0.6115
p=0.4336
p=0.3651
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

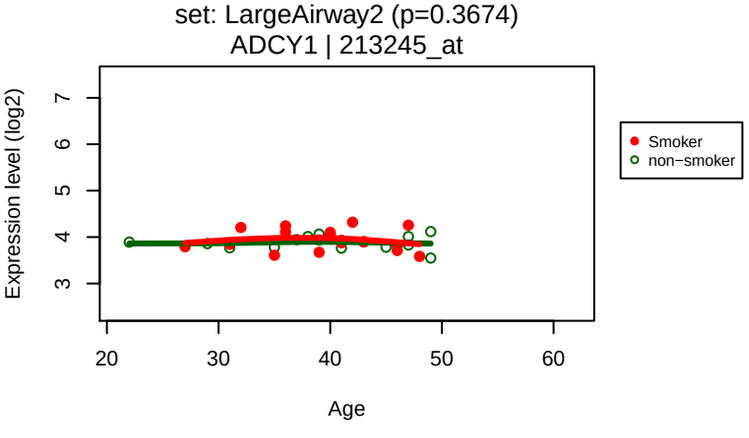

Gene expression ratio (smokers/nonsmokers)
gene: ADCY1 | 213245_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4316
p=0.9019
p=0.653
p=0.026
p=0.3674
p=0.6115
p=0.4336
p=0.3651
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

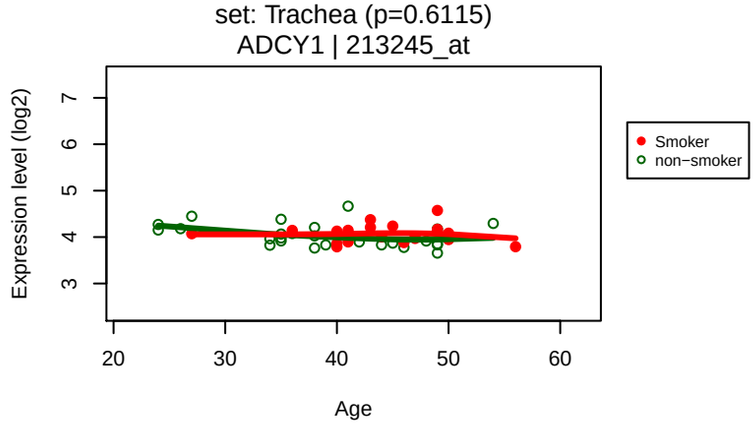

Gene expression ratio in Males
gene: ADCY1 | 213245_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9792
p=0.8139
p=0.2169
p=0.5149
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ADCY1 | 213245_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4021
p=0.2749
p=0.6692
p=0.5656
SmallAirway2
LargeAirway2
Trachea
Macrophage2

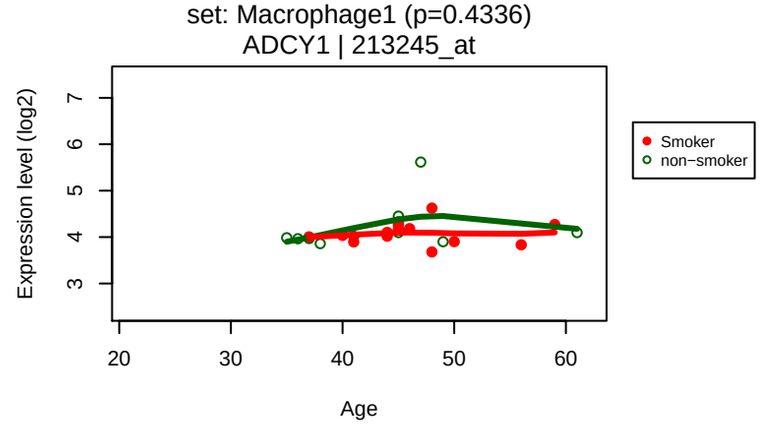

set: SmallAirway1 (p=0.4316)
ADCY1 | 213245_at
Expression level (log2)
Age
Smoker
non−smoker

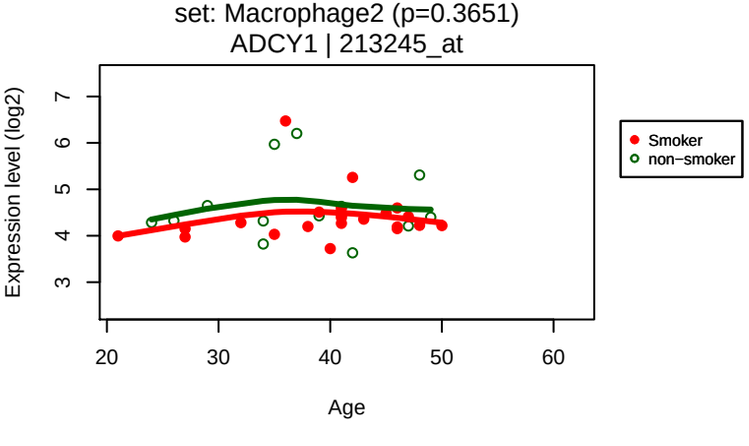

set: SmallAirway2 (p=0.9019)
ADCY1 | 213245_at
Expression level (log2)
Age
Smoker
non−smoker

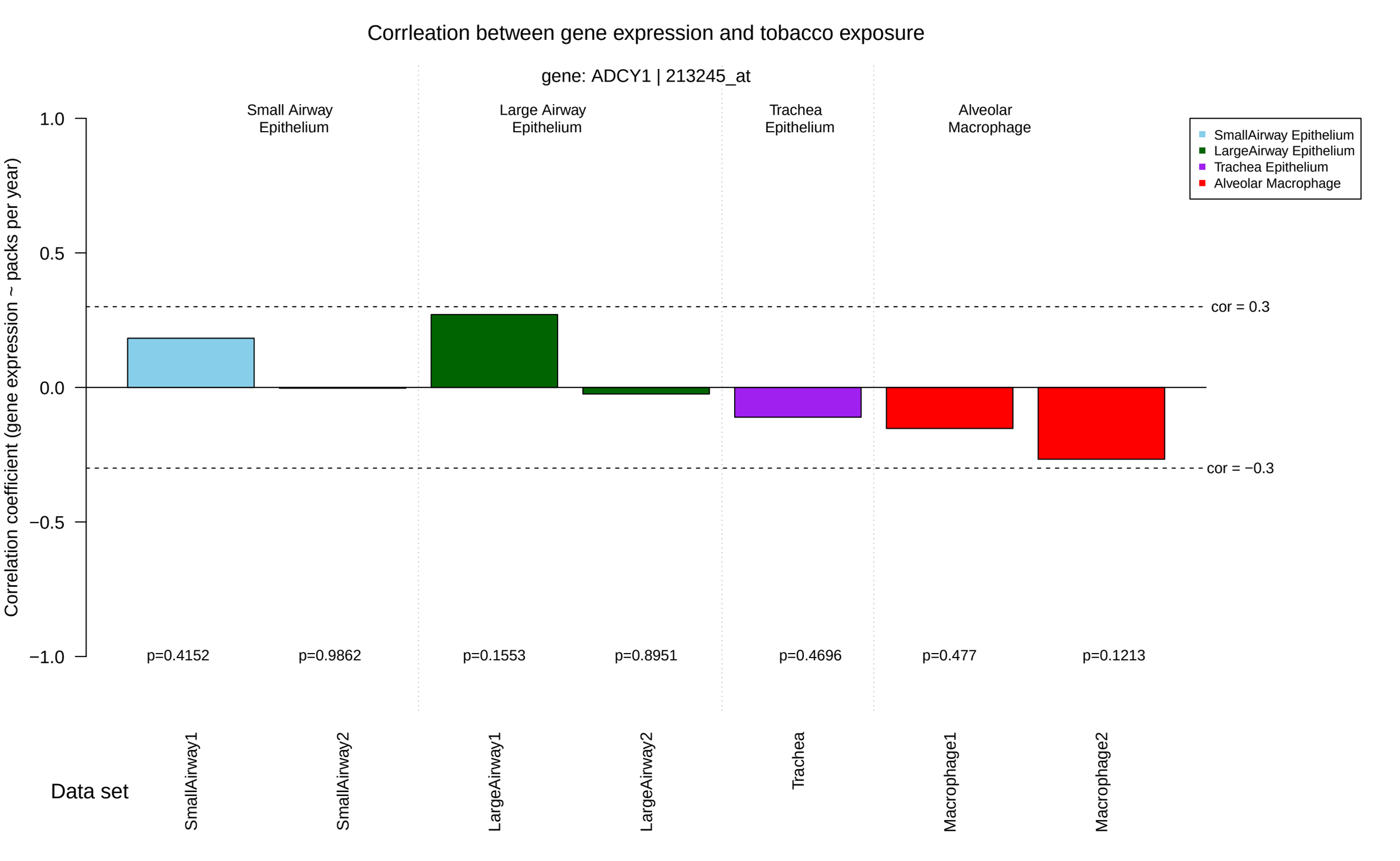

set: LargeAirway1 (p=0.026)
ADCY1 | 213245_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3674)
ADCY1 | 213245_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.6115)
ADCY1 | 213245_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.4336)
ADCY1 | 213245_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3651)
ADCY1 | 213245_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ADCY1 | 213245_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.4152
p=0.9862
p=0.1553
p=0.8951
p=0.4696
p=0.477
p=0.1213
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2